

NEUROLOGICAL EMERGENCIES

Stroke | Seizure | Meningitis | Head Injury | Loss of Consciousness

Clinical Guideline for AI Model Training Data — Version 1

Sources: WHO Stroke Guidelines | MSF Clinical Guidelines | WHO mhGAP | Hesperian Health Guides | ICRC War Surgery

Neurological emergencies require rapid recognition and early action. In conflict settings where CT imaging and specialist care are unavailable, clinical recognition and appropriate first aid can be the difference between survival with good function and death or permanent disability.

1. STROKE (CEREBROVASCULAR ACCIDENT — CVA)

1.1 FAST Recognition Tool

Letter	Test	Positive Finding
F — Face	Ask person to smile	One side of face droops — facial asymmetry
A — Arms	Ask person to raise both arms with eyes closed	One arm drifts downward or cannot be raised
S — Speech	Ask person to repeat a simple sentence	Slurred, wrong words, or unable to speak
T — Time	If ANY of the above: Time is CRITICAL	Note exact time symptoms began — critical for treatment window

1.2 Ischaemic vs Haemorrhagic Stroke — Field Differentiation

Clinical differentiation between ischaemic (clot) and haemorrhagic (bleed) stroke is difficult without imaging. Treat as stroke and refer urgently.

Feature	Ischaemic Stroke (more common)	Haemorrhagic Stroke
Onset	Often on waking or gradual	Sudden during activity
Headache	Absent or mild	Severe 'thunderclap' headache
Vomiting	Less common	More common
Consciousness	Often preserved early	May be impaired early
Progression	May stabilise or improve	Often deteriorates

1.3 Management

STEP 1	NOTE EXACT TIME
	Record time of symptom onset — critical for determining eligibility for thrombolysis if available.
STEP 2	POSITION
	Conscious patient: lie flat with head slightly elevated (20–30 degrees). Do NOT raise head above 30 degrees in ischaemic stroke — reduces cerebral perfusion.
STEP 3	AIRWAY
	Unconscious: recovery position; ensure airway is clear of secretions.

STEP 4	DO NOT GIVE FOOD OR WATER
	Swallowing impairment is common after stroke — high aspiration risk. Test swallow before any oral intake.
STEP 5	DO NOT LOWER BLOOD PRESSURE AGGRESSIVELY
	Elevated BP after stroke is a compensatory response. Do NOT treat high BP acutely in the field unless > 220/120.
STEP 6	URGENT REFERRAL
	Stroke is a time-critical emergency. Transport immediately. Stroke is not treatable in the field.

DO NOT give aspirin until haemorrhagic stroke is excluded — aspirin worsens a bleed.

DO NOT give food or fluids until swallowing is formally assessed — aspiration pneumonia risk.

Time to thrombolysis (within 4.5 hours of onset) is critical where available — do not delay transport.

1.4 Transient Ischaemic Attack (TIA — Mini Stroke)

- Same FAST symptoms as stroke but resolve completely within 24 hours (usually < 1 hour).
- TIA is a warning — 10–15% risk of stroke within 48 hours.
- Refer urgently even if symptoms have resolved completely.
- Do NOT reassure patient that 'it passed' — TIA requires urgent investigation.

2. MENINGITIS

2.1 Recognition — The Classic Triad

Sign	Description	How to Test
Severe headache	Worst headache of patient's life; often sudden onset; global	Ask patient: 'is this the worst headache you have ever had?'
Neck stiffness	Inability to flex neck; chin cannot touch chest; pain on attempted flexion	Gently flex neck forward — resistance and pain = positive
Photophobia	Eyes very sensitive to light — patient turns away from light or covers eyes	Ask patient to look at a light source
Kernig's sign	Flex hip to 90 degrees; attempt to extend knee — pain and resistance = positive	Sensitivity 50% — absence does not exclude meningitis
Brudzinski's sign	Passive neck flexion causes involuntary flexion of both knees	Test while patient lies flat
Non-blanching rash	Petechiae or purpura (small red/purple spots that do NOT blanch on pressure)	Glass test: press glass on rash — does NOT fade = meningococcal emergency

2.2 Non-Blanching Rash — Meningococcal Septicaemia

Non-blanching petechial or purpuric rash + fever = meningococcal septicaemia until proven otherwise.

This is a LIFE-THREATENING emergency — mortality can be > 20% even with treatment.

Administer IV/IM benzylpenicillin or ceftriaxone IMMEDIATELY if available — do NOT wait for confirmation.

Urgent referral — do NOT delay transport for any investigation.

2.3 Management

STEP 1	ISOLATE THE PATIENT
	Bacterial meningitis is potentially transmissible by respiratory droplets.

STEP 2	AIRWAY AND POSITIONING
	If conscious: darkened, quiet room; head elevated 20–30 degrees. If unconscious: recovery position.
STEP 3	ANTIBIOTICS
	Do NOT delay antibiotics waiting for investigations. Administer per MSF/WHO protocol immediately on clinical diagnosis.
STEP 4	FEVER MANAGEMENT
	Cooling cloths; paracetamol if available.
STEP 5	URGENT REFERRAL
	All suspected meningitis requires inpatient care. Transport immediately after first antibiotic dose.

3. HEAD INJURY

3.1 Severity Classification

Severity	GCS Score	LOC Duration	Clinical Features
Mild (concussion)	13–15	< 30 min	Brief confusion; amnesia; headache; nausea; no neurological deficit
Moderate	9–12	30 min–6 hours	Confusion, altered behaviour, possible neurological signs
Severe	3–8	> 6 hours	Unconscious, cannot follow commands, focal neurological deficits

3.2 Glasgow Coma Scale — Field Assessment

Component	Response	Score
Eye opening	Spontaneous / To voice / To pain / None	4 / 3 / 2 / 1
Verbal response	Oriented / Confused / Words only / Sounds / None	5 / 4 / 3 / 2 / 1
Motor response	Obeys commands / Localises pain / Withdraws / Flexion / Extension / None	6 / 5 / 4 / 3 / 2 / 1

Total GCS: 15 = normal; 13–14 = mild impairment; 9–12 = moderate; < 9 = severe; 3 = no function.

3.3 Dangerous Signs — Escalation Criteria

Sign	Clinical Significance
Unequal pupils (one dilated)	Possible uncal herniation — pressure on brainstem. Neurosurgical emergency.
Deteriorating GCS	Worsening consciousness = expanding intracranial haemorrhage
Seizure after head injury	Post-traumatic seizure — monitor closely; may recur
Cushing's triad	Rising BP + slowing pulse + irregular breathing = severe raised intracranial pressure
Clear fluid from nose or ear	Cerebrospinal fluid leak — basal skull fracture
Battle's sign / Raccoon eyes	Bruising behind ear or around eyes = basal skull fracture

3.4 Management

STEP 1	SPINAL PRECAUTIONS
	Assume cervical spine injury in any significant head trauma. Immobilise spine.

STEP 2	AIRWAY
	Head-tilt chin-lift (jaw thrust in trauma). Recovery position if unconscious. Suction secretions.
STEP 3	HEAD ELEVATION
	Elevate head of bed 20–30 degrees to reduce intracranial pressure. Do NOT if spinal injury not excluded.
STEP 4	AVOID SECONDARY INSULTS
	Prevent hypoxia, hypotension, fever, and hypoglycaemia — all worsen outcome.
STEP 5	MONITORING
	GCS every 30 minutes. Any deterioration = urgent referral or re-escalate care.
STEP 6	REFER
	Any GCS < 13, unequal pupils, or deteriorating GCS — neurosurgical referral.

4. LOSS OF CONSCIOUSNESS — DIFFERENTIAL APPROACH

Cause	Key Distinguishing Features	Priority Action
Vasovagal (faint)	Preceded by nausea, pallor, standing/heat/fear; rapid recovery; short duration	Lay flat; elevate legs; recovers fully
Hypoglycaemia	Known diabetic; pale; sweating; rapid onset; responds to sugar	Sugar immediately if swallowing safe
Seizure	Convulsions; tongue bite; urinary incontinence; prolonged post-ictal confusion	Recovery position; airway; time it
Stroke	Facial droop; arm weakness; speech slurred; does NOT resolve quickly	Urgent referral; recovery position
Head injury	Trauma history; scalp wound; deteriorating GCS	Spinal precautions; urgent referral
Meningitis	Fever + headache + neck stiffness + rash	Antibiotics immediately; urgent referral
Overdose / poison	Pinpoint pupils (opioid); slow breathing; history of exposure	Airway; antidote if available; referral
Cardiac cause	Preceded by chest pain or palpitations; irregular pulse; rapid recovery	Urgent cardiac referral

5. RED FLAGS — CRITERIA FOR URGENT REFERRAL

Red Flag Criterion	Clinical Significance / Action
FAST positive	Possible stroke — time-critical emergency
Non-blanching rash + fever	Meningococcal septicaemia — antibiotics immediately; urgent referral
GCS < 13 or deteriorating GCS	Severe head injury or expanding intracranial lesion
Unequal pupils after head injury	Uncal herniation — neurosurgical emergency
Status epilepticus (> 5 min)	Life-threatening — emergency management required
Thunderclap headache	Possible subarachnoid haemorrhage — urgent referral
Meningism (stiff neck + fever)	Bacterial meningitis — antibiotics and urgent referral
CSF from nose/ear	Basal skull fracture — neurosurgical referral

REF. REFERENCES

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